Yuan Jack Yao

404-528-6519 | yyao411@gatech.edu | LinkedIn | GitHub | Portfolio

Education

Georgia Institute of Technology **Aug 2025 – Dec 2026**
***School of Computational Science and Engineering**, M.S. in Analytics* *Atlanta, Georgia*

- **Graduate Teaching Assistant @ CS 6604 Conversational AI**
- Courses: Conversational AI, Natural Language, Machine Learning, ML in Computational Biology

National Taiwan University **Sep 2014 – Jun 2018**
Bachelor's Degree in Economics *Taipei, Taiwan*

Work Experiences

Walmart **Jun 2026 – Aug 2026**
Data Scientist III Intern @ Sam's Club AI Acceleration Team *Bentonville, Arkansas*

- Integrated **LLM & RAG with filtered vector search** into dashboard for answering camera metadata queries at camera/area/site levels, with **cosine similarity & LLMs-as-a-Judge as quality gates against hallucinations**
- Designed an **embedding signature to verify Milvus vector database** ingestion & retrieval sides call the identical embedding model — despite being forced onto different API clients — **eliminating silent retrieval degradation**
- Deployed **monthly Gemini Vision CronJob** across **1,245 in-store cameras** spanning 9 sites to classify snapshot content and map camera to its actual in-store area, with SVG floor layouts as primary reference and CSV as fallback
- Achieved full-fleet **inference of 1,245 cameras in 13 minutes through concurrency**, while **capping glibc malloc** arenas and **pre-downsampling images** to **eliminate out-of-memory** failures
- Deployed **hourly CronJob** with **Docker** & **Kubernetes**, integrating **asynchronous processing** that reduced **50%** execution time for capturing **1,245 camera snapshots** into **Azure**, using 7-day TTL to save **75%** storage cost

SUEZ **May 2022 – Sep 2024**
Junior Data Scientist @ Smart and Environmental Solutions (SES) China *Shanghai, China*

- Streamlined **NLP classifier** for **client's imbalanced** SCADA alarms, reaching **82% production accuracy**
- Designed a **DBSCAN anomaly detector** for **client's** air compressors predictive maintenance, achieving **70% production F1 score** and forecasting failures **two weeks in advance despite having only a single feature**

Personal Projects

LinesShines — Interactive NFL Stats System, **iOS/Android/macOS/Windows Installable**

- Designed **pinnable multi-merging** player comparison system with live-recomputing rank & percentile panels across multi-team/player union filters, inspired by GitHub's pinned repos & pull requests merge mechanics
- Architected teammate-strength aggregation by relative snap-share weighting to reflect true unit-level performance

MonkeyJOB: Jack's Online Blackjack — **Installable Online Game**

- Developed **full-stack Blackjack** with casino-style UI via FastAPI backend & vanilla JavaScript
- Implemented Macau-style split logic supporting up to 4 split hands, with conditional double-after-split eligibility

SkyHorse — Bilingual Tech Blog

- Authored **30+ LeetCode hard problems** tutorials to explain step-by-step Python & C++ implementations

Open Source

Pydantic AI (20k Stars, 2.7k Forks) — Agentic AI Framework

- **Unified API errors handling pattern** across Google & Cohere embedding models by shared context managers

Code Puppy (816 Stars, 280 Forks) — Coding Agents

- **Cut down idle CPU wake-ups by 88.5% for Agent-UI communications, restored silently-dropped user config across Gemini classes**, unified tools' drifted file-read process, and aligned 20+ inconsistent parser calls

LeetCode Wiki (36.6k Stars, 9.5k Forks) — DSA Library

- Authored **8 optimal time and space solutions with tutorials** for **hard** problems
- **No.2454 Next Greater Element IV** optimization: achieved O(n) time complexity by **double stacks**

Technical Skills

Gen AI & ML: AI Infra & Platform, RAG, LLM Fine-tuning, Distillation, PyTorch, Hugging Face, TensorFlow, Keras
CI/CD & Cloud: Linux, GitHub, Bitbucket, Docker, Kubernetes, Azure, AWS, GCP
Programming & Databases: Python, C++, Milvus, Cassandra, MySQL, PostgreSQL, ETL Pipelines
Web & Visualization: FastAPI, Flask, JavaScript, HTML, CSS, Web Scraping